

Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

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🌐 linkedin.com/in/amir-golpayegani-218148165 | 🐙 github.com/amirgolpaa24 | 🌐 Portfolio

Summary

M.Sc. Computer Science graduate with experience in Python-based machine learning, statistical analysis, model evaluation, and large-scale data workflows. Skilled in PyTorch, SQL-backed backend systems, data pipelines, and scientific computing, with hands-on experience turning ambiguous research problems into reproducible analyses, models, and clear technical recommendations.

Education

M.Sc. in Computer Science University of Calgary	GPA: 3.94/4.0	Sep 2022 – Feb 2026 Calgary, Canada
B.Sc. in Computer Engineering Sharif University of Technology	GPA: 3.69/4.0	Sep 2016 – Feb 2021 Tehran, Iran

Technical Skills

Programming / Data: Python, SQL, C++, JavaScript, Git
ML / AI: PyTorch, Scikit-learn, TensorFlow, CNNs, predictive modeling, model evaluation, LLM workflows, AI agents
Analysis / Experimentation: NumPy, Pandas, SciPy, feature engineering, validation design, statistical metrics, reproducible research
Data / Geospatial: Google Earth Engine API, Discrete Global Grid System (DGGS), GDAL, Rasterio, SNAP, satellite imagery
Backend / Engineering: Django, Node.js, REST APIs, PostgreSQL, Docker, Linux, AWS, GitHub Actions
Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

Work Experience

Research Software Developer Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary Calgary, Canada

- Built C++/Python software for scalable processing, storage, retrieval, and analysis of Earth observation data.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert satellite imagery into structured tensor files for downstream analysis.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGS tensors.

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz Tehran, Iran

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

Deep Learning Teaching Assistant Jul 2026
Neuromatch Academy Remote, United States

- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary Calgary, Canada

- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** +75 Assisted students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

DEM Super-Resolution Framework May 2024 – Feb 2026
Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Rasterio, Model Evaluation

- Trained deep ResNet models for $4\times$ DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built Python pipelines to collect, preprocess, and validate mountainous terrain data from Google Earth Engine.
- Evaluated models using RMSE, slope, TRI, TPI, wavelet-based errors, and visual 3D inspection of reconstruction quality.
- Identified that random data splitting produced overly optimistic results and designed region-based validation for stronger real-world assessment.

Spatiotemporal Earth Observation Data System Using DGGs

Sep 2022 – Feb 2026

[First-author publication](#) | [M.Sc. thesis](#) | C++, Python, DGGs, Wavelets, B-splines, Time-Series

- Designed a DGGs-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.
- Built B-spline temporal approximation to compress time-series data and retrieve missing timestamps locally.
- Retrieved 105,489 DGGs cells per time-series query in 13.9s, compared with about 220s using global retrieval.

Agentic Job Search Automation Platform

May 2026 – Present

OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API

- Building an agentic AI system to search jobs, analyze fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and human-reviewable application decisions.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.

Persian Text Normalizer for NLP

Oct 2016 – Feb 2021

Python, NLP, Text Processing

- Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines.
- Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs*. Remote Sensing, 2025. [doi:10.3390/rs17040570](#)

Certificates

Supervised Machine Learning: Regression and Classification , DeepLearning.AI and Stanford University, Coursera	Oct 2024
Applied Machine Learning in Python , University of Michigan, Coursera	Jul 2020

Languages

English: Full Professional Proficiency (TOEFL 115/120)
Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate Department of Computer Science, University of Calgary — awarded for outstanding teaching support (link)	2024/2025
Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) Top 0.001% among all participants	May 2016